Citation for published version:
Wortham, RH, Theodorou, A & Bryson, JJ 2016, What Does the Robot Think? Transparency as a Fundamental Design Requirement for Intelligent Systems. in *Proceedings of the IJCAI Workshop on Ethics for Artificial Intelligence : International Joint Conference on Artificial Intelligence.* IJCAI 2016 Ethics for AI Workshop, New York, USA United States, 9/07/16.

Publication date:
2016

Document Version
Publisher's PDF, also known as Version of record

Link to publication

Publisher Rights
Unspecified

University of Bath

General rights
Copyright and moral rights for the publications made accessible in the public portal are retained by the authors and/or other copyright owners and it is a condition of accessing publications that users recognise and abide by the legal requirements associated with these rights.

Take down policy
If you believe that this document breaches copyright please contact us providing details, and we will remove access to the work immediately and investigate your claim.

Download date: 08. Sep. 2019

What Does the Robot Think? Transparency as a Fundamental Design Requirement for Intelligent Systems

Robert H. Wortham, Andreas Theodorou, Joanna J. Bryson
Department of Computer Science, University of Bath, UK
r.h.wortham@bath.ac.uk, a.theodorou@bath.ac.uk, jjb@alum.mit.edu

Abstract

Deciphering the behaviour of intelligent others is a fundamental characteristic of our own intelligence. As we interact with complex intelligent artefacts, humans inevitably construct mental models to understand and predict their behaviour. If these models are incorrect or inadequate, we run the risk of self deception or even harm. This paper reports progress on a programme of work investigating approaches for implementing robot transparency, and the effects of these approaches on utility, trust and the perception of agency. Preliminary findings indicate that building transparency into robot action selection can help users build a more accurate understanding of the robot.

1 INTRODUCTION

Article four of the EPSRC Principles of Robotics asserts that *Robots are manufactured artefacts. They should not be designed in a deceptive way to exploit vulnerable users; instead their machine nature should be transparent.* [1]. Why is Transparency important, and how does it impact AI system design? Writers such as Mueller [11] suggest that as intelligent systems become both increasingly complex and ubiquitous, it becomes increasingly important that they are self explanatory, so that users can be confident about what these systems are doing and why. Mueller sees explanation as one of the three main characteristics of transparent computers, the others being dialogue and learning.

Humans have a natural if limited ability to understand others, however this ability has evolved and developed in the environment of human and other animal agency, which may make assumptions artificial intelligence does not conform to. Therefore it is the responsibility of the designers of intelligent systems to make their products transparent to us [19; 15]. This is of particular importance when deploying robots in environments where those who interact with them may be vulnerable, such as in care homes or hospitals [14], or equally in high-risk environments where misunderstanding a robot may have dangerous consequences.

Note that the need to form a useful model is orthogonal to issues of verification of robot behaviour. Whilst others have concentrated their research on making a robot safe and predictable [8; 16], we are interested here in the models that observers of a robot use to understand and predict its behaviour.

Decoding the behaviour of intelligent others is a fundamental characteristic of our own intelligence. It is generally thought that many forms of effective interaction, whether coercion or co-operation, rely on each party having some theory of mind (ToM) concerning the other [17; 13]. Individual actions and more complex behaviour patterns are thus interpreted within a pre-existing ToM framework. Whether that ToM is entirely accurate is unimportant, provided that it is predictive in terms of behaviour. Humans have a strong tendency to anthropomorphise not only nature, but anything around them [6] — the Social Brain Hypothesis [7] may explain this phenomenon. As we interact with complex intelligent artefacts, we construct anthropomorphic models to understand and predict their behaviour. If these models are incorrect, or inadequate, we are at best at risk of being deceived and at worse at risk of being harmed.

This paper reports preliminary findings from human-subject experiments concerning the understanding of a simple autonomous robot. Subjects watch a video of a robot interacting with a researcher, and report their theories about what the robot is doing and why. Some of these reports are wildly inaccurate, and interestingly many conclude that the robot's objectives and abilities are far more complex than they in fact are. Importantly, we find that simply showing the runtime activation of the robot's action selection along with the video results in users building significantly more accurate models of the robot's behaviour.

2 BACKGROUND: REACTIVE PLANNING & ROBOT TRANSPARENCY

Here we use reactive planning techniques to build transparent autonomous agents. We have deployed the *Instinct* reactive planner [18] as the core action selection mechanism for the R5 robot. Instinct is deployed in the context of Bryson's Behaviour Oriented Design (BOD) development methodology, as a replacement and extension of Parallel-rooted, Ordered Slip-stack Hierarchical (POSH) action selection [1] [3]. Instinct includes several enhancements taken from more recent papers extending POSH [12; 9], together with some ideas

[1] POSH — http://www.cs.bath.ac.uk/∼jjb/web/posh.html

from other related planning approaches, notably Behaviour Trees (BT) [10]. A POSH plan consists of a *Drive Collection (DC)* containing one or more *Drives*. Each Drive (D) has a priority and a releaser. When the *Drive* is released as a result of sensory input, a hierarchical plan of *Competences*, *Action Patterns* and *Actions* follows.

The Instinct Planner has been specifically designed for low-power processors and has a tiny memory footprint. Written in C++, it runs efficiently on both ARDUINO (ATMEL AVR) and MICROSOFT VC++ environments and has been deployed within a low-cost ARDUINO-based maker robot for this study of AI transparency. We have named this robot R5, in reference to the Rover 5 tracked platform on which it is based. Plans may be authored using a variety of tools including a promising visual design language *iVDL*, currently implemented using the DIA drawing package. The Instinct Planner and iVDL are available on an open source basis[2].

2.1 Robot Plans

POSH plans are written in a LISP like notation, either using a text editor, or the Advanced Behaviour Oriented Design Environment (ABODE) [3] editor [2], which allows graphical representations of the plans.

However, Instinct plans are written very differently, because they must use a much more compact notation due to memory constraints. We make use of the *Instinct Visual Design Language (iVDL)* — a graphical method of designing reactive plans, based on the ubiquitous Unified Modelling Language (UML) notation. UML is supported by many drawing packages and a simple PYTHON export script provided as part of Instinct allows plans to be created graphically within the DIA[4] drawing tool. An example robot plan is shown in Figure 1. At this level of zoom the element details are not legible, but this screen shot gives an impression of how plans can be laid out, and the complexity of the plan used in our experiment.

2.2 The Transparent Planner

The Instinct Planner includes significant capabilities to facilitate plan design and runtime debugging. It reports the execution and status of every plan element in real time, allowing us to implicitly capture the reasoning process within the robot that gives rise to its behaviour. The planner has the ability to report its activity as it runs, by means of callback functions to to a monitor class. There are six separate callbacks monitoring the Execution, Success, Failure, Error and In-Progress status events, and the Sense activity of each plan element. In the R5 robot, the callbacks write textual data to a TCP/IP stream over a wireless (wifi) link. A JAVA based Instinct Server receives this information and logs the data to disk. This communication channel also allows for commands to be sent to the robot while it is running. Figure 2 shows the overall architecture of the planner within the R5 robot, communicating via wifi to the logging server.

[2] http://www.robwortham.com/instinct-planner/

[3] http://www.cs.bath.ac.uk/∼jjb/web/BOD/abode.html

[4] Dia — http://dia-installer.de/

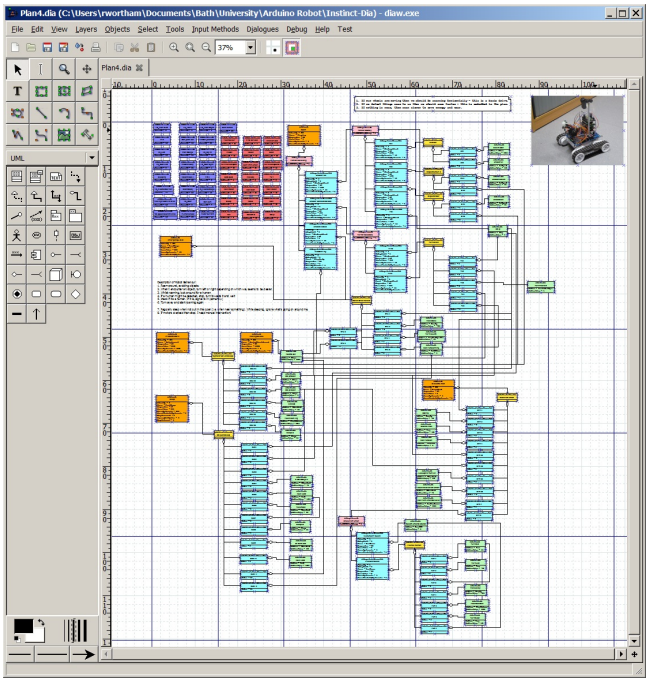

Figure 1: The Robot Plan Developed in iVDL

2.3 Realtime Plan Debugger

We are fortunate to have access to a new pre-alpha version of the ABODE plan editor, *ABOD3*, as seen in Figure 3. This version directly reads Instinct plans, and also reads a log file containing the real-time transparency data emanating from the Instinct Planner, in order to provide a real-time graphical display of plan execution. ABOD3 is also able to display a video and synchronise it with the debug display. In this way we can explore both runtime debugging and wider issues of AI Transparency.

3 METHODS: THE ROBOT EXPERIMENT

The robot in the video runs within an enclosed environment where it interacts with various objects and walls made of different materials. A researcher also interacts with the robot. The robot's action selection governs the behaviour of the robot by applying the reactive plan. A reactive plan encodes the priorities of a robot and the conditions when actions can be applied. A record of transparency data in the form of a log of which plan components are triggered at what time is collected by a remote server running on a laptop PC via a wifi connection.

Using its built-in real time clock, the robot tags the transparency datastream with the start time of the experiment. It also includes the elapsed time in milliseconds with every datastream event. In this way the ABOD3 debugger is able to subsequently synchronise the datastream with video recordings taken during the experiment.

3.1 Robot Drives and Behaviours

The robot plan shown in Figure 1 has six Drives. These are (in order of highest priority first):

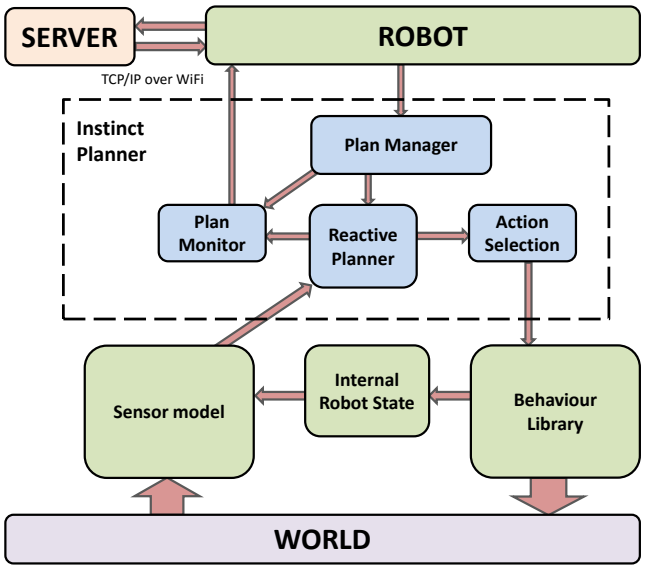

Figure 2: R5 Robot Software Architecture

- Sleep — this Drive has a ramping priority. Initially the priority is very low but it increases linearly over time until the Drive is released and completes successfully. The Drive is only released when the robot is close to an obstacle. This is to prevent the robot sleeping in the middle of an open space where it may present a trip hazard. The sleep behaviour simply shuts down the robot for a fixed interval to conserve battery power.
- Protect Motors — released when the current drawn by the drive motors reaches a threshold. This might happen if the robot encounters a very steep incline or becomes jammed somehow. The Drive invokes an Action Pattern that stops the robot, signals for help and then pauses to await assistance.
- Moving So Look — simply enforces that if the robot is moving, it should be scanning ahead for obstacles. This has a high priority so that this rule is always enforced whatever else the robot may be doing.
- Detect Human — released when the robot has moved a certain distance from its last confirmed detection of a human, is within a certain distance of an obstacle ahead, and its Passive Infrared (PIR) detects heat that could be from a human. This Drive initiates a fairly complex behaviour of movement and coloured lights designed to encourage a human to move around in front of the robot. This continues to activate the PIR sensor thus confirming the presence of a human (or animal). It is of course not a particularly accurate method of human detection.
- Emergency Avoid — released when the robot's active infrared corner sensors detect reflected infrared light from a near obstacle. This invokes a behaviour that reverses the robot a small distance and turns left or right a fixed number of degrees. Whether to turn left or right is determined by which direction appears to be less blocked, as sensed by the active infrared detectors.
- Roam — released whenever the robot is not doing anything else. It uses the scanning ultrasonic detector to determine when there may be obstacles ahead and turns appropriately to avoid them. It also modulates the robot speed and the rate of scanning depending on the proximity of obstacles.

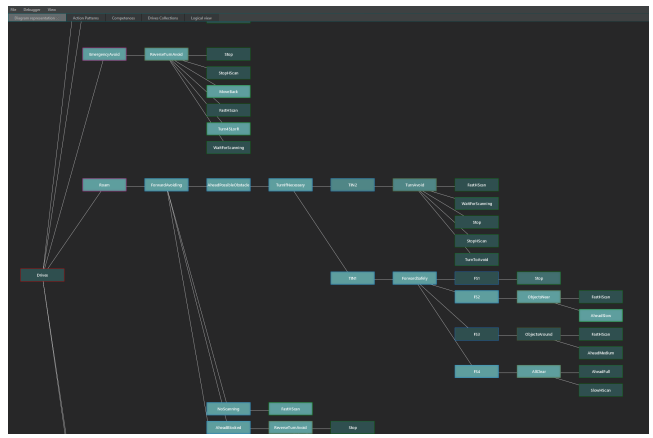

Figure 3: ABOD3 Showing Part of the Instinct Plan

3.2 Robot Videos

For our pilot study, we chose to video the robot rather than have participants interact with the robot directly. This research method has recently been chosen by others [4] with good results. Video has the benefit of ensuring all subjects share identical stimuli.

Figure 4: Video of interaction with the robot with no plan visible (stimulus for Group One).

The interaction is recorded from two positions at each end of the robot pen, and a camera mounted on a post attached to the robot also captures a 'robots eye' view, providing a third perspective. The resulting composite video is approximately five minutes long. Figure 4 is a single frame from the video. It shows the researcher interacting with the robot. This video was shown to half of our group of test participants.

Using the pre-alpha version of the ABOD3 tool, we created a second video. A frame from this video is shown in Figure 5. The six Drives described above are clearly visible. As each Drive is released and the associated behaviours are executed, the plan elements constituting the behaviours are highlighted. This highlighting is synchronised with the behaviour of the robot visible in the video. This gives the viewer access to a great deal more information from the robot than is available by watching the robot alone. ABOD3 conveniently allows us to collapse the lower levels in the hierarchy, and position the visible plan elements for ease of understanding. For the purpose of clarity in the video, we chose to display only the highest levels of the reactive plan, primarily the robot Drives.

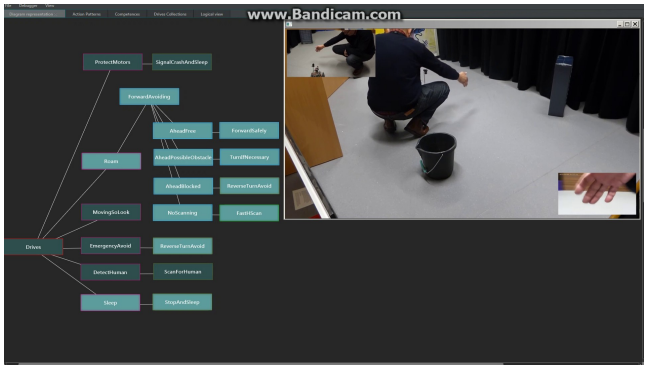

Figure 5: More transparent video showing the ABOD3 plan representation; sub-trees have been hidden from view (stimulus for Group Two).

3.3 Demographic & Post-Treatment Questionnaires

The participants were initially sent an email questionnaire to gather basic demographic data: age, gender, educational level, whether they use computers, whether they program computers and whether they have ever used a robot. Based on this information they were then divided into two groups that were matched as nearly as possible for participant mix. Each group received an identical email asking them to carefully watch a video and then answer a second questionnaire. Group One was directed to the composite video (Fig 4), and Group Two to the debug video (Fig 5).

Table 1 summarises the questions asked after the participant had seen the video. These questions are designed to measure various factors: the measure of intelligence perceived by the participants (Intel), the emotional response (if any) to the robot (Emo), and—most importantly—the accuracy of the participants' mental model of the robot (MM). For analysis, the four free text responses were rated for accuracy with the robot's actual Drives & behaviours and given a score per question of 0 (inaccurate or no response), 1 (partially accurate) or 2 (accurate). Question 3 was found to be ambiguous and so is not included in the scores, see below. By summing the scores, the accuracy of the participant's overall mental model is scored from 0 to 6.

Table 1: Post-Treatment Questions

Question	Response	Category
Is robot thinking?	Y/N	Intel
Is robot intelligent?	1-5	Intel
Feeling about robot?	Multi choice	Emo
Understand objective?	Y/N	MM
Describe robot task?	Free text	MM
Why does robot stop?	Free text	MM
Why do lights flash?	Free text	MM
What is person doing?	Free text	MM
Happy to be person?	Y/N	Emo
Want robot in home?	Y/N	Emo

4 RESULTS

The demographics of each group of participants is shown in Table 3. Priority was given to matching the number of programmers in each group, and to having an equal gender mix.

Table 2: Main Results. Bold face indicates results significant to at least $p = .05$.

Result	Group One	Group Two
Is thinking (0/1)	0.36 (sd=0.48)	0.65 (sd=0.48)
Intelligence (1-5)	2.64 (sd=0.88)	2.74 (sd=1.07)
Undrstnd objctv (0/1)	0.68 (sd=0.47)	0.74 (sd=0.44)
Rpt Accuracy (0-6)	1.86 (sd=1.42)	3.39 (sd=2.08)

Table 3: Demographics of Participant Groups $(N = 45)$

Demographic	Group One	Group Two
Mean Age (yrs)	39.7	35.8
Gender Male	11	10
Gender Female	11	12
Gender PNTS	0	1
Total Participants	22	23
STEM Degree	7	8
Other Degree	13	13
Ever worked with a robot?	2	3
Do you use computers?	19	23
Are you a Programmer?	6	8

4.1 Main Findings

The primary results obtained from the experiment are outlined in Table 2. Firstly and most importantly, there is a marked difference in the participants' mental model accuracy scores between Group One (video only; m=1.86, sd=1.42) and Group Two (video plus ABOD3 debug display; m=3.39, sd=2.08); t(43)=2.86, p=0.0065, d=0.53. This confirms a significant correlation between the accuracy of the participants' mental models of the robot, and the provision of the additional transparency data provided by ABOD3 in the video. Secondly, there is no significant difference in perceived robot intelligence between the two groups, with Group

One reporting an average intelligence rating of 2.64 (sd=0.88) and Group Two reporting 2.74 (sd=1.07); t(43)=0.35 p=0.73 d=0.29. However, a substantially higher number of participants in Group Two (ABOD3) report that they believe the robot is thinking; t(43)=2.02, p=0.050.

4.2 Qualitative Outcomes

The data indicate very little emotional response to the robot in either group, with most participants indicating either 'No Feeling', or 'Curious'.

From the answers to the question 'why does the robot stop every so often' it appears that this question is ambiguous. Some understand this to mean every time the robot stops to scan its environment before proceeding, and only one person took this to mean the sleep behaviour of the robot that results in a more prolonged period of inactivity. The question was intended to refer to the latter, and was particularly included because the Sleep Drive is highlighted by ABOD3 each time the robot is motionless with no lights flashing. However only one member of Group Two identified this from the video. Due to this ambiguity, the data related to this question was not considered further.

Despite the improved performance of Group Two, many members, even those with a Science, Technology Engineering or Maths (STEM) degrees, still form a poor mental model of the robot. Here are some notable quotes from Group Two STEM participants:

- [the robot is] Trying to create a 3d map of the area? At one stage I thought it might be going to throw something into the bucket once it had mapped out but couldn't quite tell if it had anything to throw.
- [the robot is] aiming for the black spot in the picture.
- is it trying to identify where the abstract picture is and how to show the complete picture?
- [the robot] is circling the room, gathering information about it with a sensor. It moves the sensor every so often in different parts of the room, so I think it is trying to gather spacial information about the room (its layout or its dimensions maybe).

5 DISCUSSION

There is a significant correlation between the accuracy of the participants' mental models of the robot, and the provision of the additional transparency data provided by ABOD3. We have shown that a real-time display of a robots decision making produces significantly better understanding of that robot's intelligence, even though that understanding may still include wildly inaccurate overestimation of the robot's abilities.

Strikingly, there was one further significant result besides the improved mental model. Subjects who observed the real-time display did not think the robot was more intelligent, but *did* think it 'thought' more. This result is counter-intuitive. We had expected that if ABOD3 resulted in increased transparency, that there would be a corresponding reduction in the use of anthropomorphic cognitive descriptions. However at least in this case the data suggests the reverse is true. When taken with the significant improvement in understanding of the robot's actual drives and behaviours, this result implies that an improved mental model is associated with an increased perception of a thinking agent. Most likely this reflects the still pervasive belief that navigating in the real world is not a difficult task, so the amount of different behaviours employed by the robot may come as a surprise.

Intelligence, however, is a term that in ordinary language is often reserved for the things we apply conscious decision making. Notably, while subjects exposed to the ABOD3 visualisations of the robot's decision making considered the robot to be thinking more, they did not consider it to be more intelligent. In fact, the middling marks for intelligence in either condition may reflect a society-wide lack of certainty about the definition of the term rather than any cognitive assessment. Indeed the relatively large standard deviations for intelligence in Table 2 provide some evidence of this uncertainty.

It might be expected that other forms of transparency display would better serve nonspecialists i.e. those not familiar with reactive planning or the ABOD3 presentation paradigm. One interesting approach would be to use argumentation techniques to generate natural language [5], although this assumes that the plan contains sufficient information to allow an explanation to be generated.

The lack of emotion with respect to the robot was unexpected, and conflicts with the spontaneous feedback we frequently receive about the robot when people encounter it in our laboratory or during demonstrations. In these situations we often hear both quite strong positive and negative emotional reactions. Some find the robot scary or creepy, whilst others remark that it is cute, particularly when it is operational. We hypothesise that the remote nature of the video, or the small size of the robot on screen, reduce the chance of significant emotional response.

6 CONCLUSION & FURTHER WORK

We have demonstrated that subjects can show marked improvement in the accuracy of their mental model of a robot observed on video, if they also see an accompanying display of the robot's real-time decision making. Although these are only the preliminary results of a small pilot study ($N = 45$), the outcome was strongly significant. The addition of ABOD3 visualisation of the robot's intelligence does indeed make the machine nature of the robot more transparent.

The results also imply that an improved mental model of the robot is associated with an increased perception of a thinking machine, even though there is no significant change in the level of perceived intelligence. The relationship between the perception of intelligence and thinking is therefore not straightforward. There is clearly further work to be done to unpack the relationship between the improved mental model of the robot and the increased perception of a thinking machine.

This research confirms that the approach of using online video with web based questionnaires is both effective and efficient in terms of researcher time, and it has enabled us to quickly gather preliminary results from which further experi-

ments can be planned. However, we did not gather any useful data about the emotional response of the participants. This may be due to possibly due to the lack of physical robot presence. If this proves true, then in situations where the emotional engagement of users to robots is of interest, the use of video techniques may prove ineffective. We intend to explore this further in future work. Further work may include generating natural language as an alternative, or addition to the ABOD3 display. This would enable us to investigate its effect on mental model accuracy.

The technology used to construct the experimental system was found to be reliable, robust and straightforward to use. Given the low cost of the platform, we would recommend its use for similar low cost research robot applications. The Instinct Planner combined with the iVDL graphical design tool enabled us to quickly generate a reliable yet sufficiently complex reactive plan for the R5 robot to allow us to conduct this experiment. Despite using the early pre-alpha version of ABOD3, it confirmed its usefulness both as a tool during robot plan debugging, and to provide transparency information to untrained observers of the the robot.

References

[1] M. Boden, J. Bryson, D. Caldwell, K. Dautenhahn, L. Edwards, S. Kember, P. Newman, V. Parry, G. Pegman, T. Rodden, T. Sorell, M. Wallis, B. Whitby, and A. Winfield. Principles of robotics. The United Kingdom's Engineering and Physical Sciences Research Council (EPSRC), April 2011. web publication.

[2] C. Brom, J. Gemrot, M. Bida, O. Burkert, S. J. Partington, and J. J. Bryson. POSH Tools for Game Agent Development by Students and Non-Programmers. In *The Nineth International Computer Games Conference: AI, Mobile, Educational and Serious Games.*, pages 1–8, Bath, UK, 2006. The University of Bath.

[3] J. J. Bryson. *Intelligence by Design: Principles of Modularity and Coordination for Engineering Complex Adaptive Agents*. PhD thesis, MIT, 2001.

[4] D. Cameron, E. C. Collins, A. Chua, S. Fernando, O. McAree, U. Martinez-Hernandez, J. M. Aitken, L. Boorman, and J. Law. Help! I Can't reach the buttons: Facilitating helping behaviors towards robots. In *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, volume 9222, pages 354–358, 2015.

[5] M. Caminada, R. Kutlak, N. Oren, and W. W. Vasconcelos. Scrutable plan enactment via argumentation and natural language generation. *Proceedings of the 2014 International Conference on Autonomous Agents and Multi-Agent Systems*, pages 1625–1626, 2014.

[6] K. Dautenhahn. Methodology & themes of human-robot interaction: A growing research field. *International Journal of Advanced Robotic Systems*, 4(1 SPEC. ISS.):103–108, 2007.

[7] R. I. M. Dunbar. The Social Brain Hypothesis. *Evolutionary Anthropology*, pages 178–190, 1998.

[8] M. Fisher, L. Dennis, and M. Webster. Verifying Autonomous Systems. *Communications of the ACM*, 56(9):84–93, 2013.

[9] S. Gaudl and J. J. Bryson. The Extended Ramp Goal Module: Low-Cost Behaviour Arbitration for Real-Time Controllers based on Biological Models of Dopamine Cells. *Computational Intelligence in Games 2014*, 2014.

[10] C. U. Lim, R. Baumgarten, and S. Colton. Evolving behaviour trees for the commercial game DEFCON. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 6024 LNCS(PART 1):100–110, 2010.

[11] E. T. Mueller. *Transparent Computers: Designing Understandable Intelligent Systems*. Erik T. Mueller, San Bernardino, CA, 2016.

[12] P. Rohlfshagen and J. J. Bryson. Flexible Latching: A Biologically-Inspired Mechanism for Improving the Management of Homeostatic Goals. *Cognitive Computation*, 2(3):230–241, 2010.

[13] R. Saxe, L. E. Schulz, and Y. V. Jiang. Reading minds versus following rules: dissociating theory of mind and executive control in the brain. *Social neuroscience*, 1(3-4):284–98, jan 2006.

[14] A. Sharkey and N. Sharkey. Granny and the robots: Ethical issues in robot care for the elderly. *Ethics and Information Technology*, 14(1):27–40, 2012.

[15] A. Theodorou, R. H. Wortham, and J. J. Bryson. Why is my Robot Behaving Like That? Designing Transparency for Real Time Inspection of Autonomous Robots. In T. J. Prescott, editor, *EPSRC Principles of Robotics Workshop, Proceedings of the AISB 2016 Annual Conference*, Sheffield, UK, 2016.

[16] A. F. T. Winfield, C. Blum, and W. Liu. Towards an Ethical Robot : Internal Models , Consequences and Ethical Action Selection. In M. Mistry, A. Leonardis, M. Witkowski, and C. Melhuish, editors, *Advances in Autonomous Robotics Systems*, pages 85–96. Springer International Publishing, Switzerland, 2014.

[17] R. H. Wortham and J. J. Bryson. Communication. In *Handbook of Living Machines* {*accepted for publication*}. Oxford University Press, Oxford, 2016.

[18] R. H. Wortham, S. E. Gaudl, and J. J. Bryson. Instinct : A Biologically Inspired Reactive Planner for Embedded Environments. In *Proceedings of ICAPS 2016 PlanRob Workshop* {*accepted for publication*}, London, UK, 2016.

[19] R. H. Wortham, A. Theodorou, and J. J. Bryson. Robot Transparency, Trust and Utility. In *EPSRC Principles of Robotics Workshop, Proceedings of the AISB 2016 Annual Conference* {*accepted for publication*}, Sheffield, UK, 2016.